

Silver Linden

Tilia tomentosa

Species background: Native to southeastern Europe and western Asia, valued as a street tree partly because it tolerates heat, drought, and urban pollution better than other lindens. Every leaf has a felted, silvery-white underside caused by dense microscopic hairs -- flip one over on a still day to see the contrast against the dark green top.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Silver Linden: Feel silver linden's fuzzy leaf underside and describe the texture.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Flip one of my leaves over -- the underside is silvery white and soft, like felt! When the wind blows, my whole canopy seems to shimmer.

Procedure

- 1 Engage: Ask students to describe something soft and fuzzy they've touched.
- 2 Explore: Gently flip a real or pictured leaf to see the silvery underside.
- 3 Explain: Explain that tiny hairs make the underside look silvery and feel soft.
- 4 Art: Feel the fuzzy underside of a leaf, then draw a fuzzy texture with dots.
- 5 Evaluate: Student can describe the leaf underside as soft, fuzzy, and silvery.

Discussion questions

What other things in nature feel soft and fuzzy?

Assessment

Correctly describes the fuzzy silvery underside.

Extension

Watch this tree's leaves shimmer in the wind if it's breezy.

GRADE BAND 3-5

Silver Linden: Compare silver linden's underside to silver maple's and identify each mechanism.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Compare this leaf's underside to a silver maple's -- which one is silvery, and why might that be? Silver linden's silvery underside comes from tiny hairs (trichomes).

Procedure

- 1 Engage: Ask if two things that look similar could work in completely different ways.
- 2 Explore: Compare a silver linden leaf and a silver maple leaf if both are available.
- 3 Explain: Explain that linden's silver comes from tiny hairs, while maple's comes from a waxy coating.
- 4 Art: Draw a two-toned leaf -- dark green marker on top, silver or white crayon on the bottom.
- 5 Evaluate: Student names the correct mechanism (hairs) for this species' silvery underside.

Discussion questions

Why might two unrelated trees end up with a similar silvery trick using totally different methods?

Assessment

Correctly identifies trichomes (hairs) as this species' silvery mechanism.

Extension

Find a silver maple nearby and compare its underside directly.

Silver Linden: Explain reflective leaf adaptations and their function in reducing water loss.

Standards connection: NGSS MS-LS4-4; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, mixed-media rubbing materials.

Teacher background

Silver linden's silvery underside comes from tiny hairs (trichomes); silver maple's comes from a waxy coating -- same visual effect, two different mechanisms.

Procedure

- 1 Engage: What environmental problem might a pale, reflective leaf underside help solve for a tree?
- 2 Explore: Research how trichomes and waxy cuticles each reduce water loss and reflect heat.
- 3 Explain: Compare the two mechanisms directly, using silver linden and silver maple as examples.
- 4 Art: Create a textured mixed-media leaf piece, using a rubbing technique for the fuzzy underside and smooth marker for the top -- write one sentence on why the two sides might feel different.
- 5 Evaluate: Write a paragraph explaining both mechanisms and their shared function.

Discussion questions

If both mechanisms solve the same problem, why do you think two unrelated trees evolved different solutions instead of the same one?

Assessment

Paragraph correctly explains both trichomes and waxy cuticle mechanisms.

Extension

Research one other tree with a documented water-conserving leaf adaptation.

Silver Linden: Evaluate convergent evolution using reflective leaf adaptations as a case study.

Standards connection: NGSS HS-LS4-4; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, hand lens or phone macro camera if available.

Teacher background

Silver linden's pale underside comes from dense trichomes, while silver maple's comes from a waxy cuticle -- two unrelated structural solutions to the same problem of reducing water loss and reflecting heat.

Research one other example of convergent evolution in plants and compare the underlying mechanisms.

Procedure

- 1 Engage: What's the technical difference between convergent evolution and simply sharing a common ancestor's trait?
- 2 Explore: Research one additional documented example of convergent evolution in plants.
- 3 Explain: Compare the trichome and waxy-cuticle mechanisms as a clear case of convergence, and contrast it with your researched example.
- 4 Art: Examine the leaf's underside with a hand lens or phone macro camera and sketch the trichome (hair) structure you observe, comparing it to a reference image of silver maple's smooth waxy underside.
- 5 Evaluate: Submit the convergent-evolution comparison and the macro-observation sketch.

Discussion questions

Does convergent evolution suggest there's often only one or two 'best' solutions to a given environmental problem?

Assessment

Rubric covering research depth and accuracy of the macro-observation sketch.

Extension

Compare this convergence case to purple leaf plum and Crimson King maple's independently evolved purple-leaf cultivars.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.